Habib University

Course Title: Cryptography and Network Security (EE-426 / CE-442)

Instructors: Dr. Farhan Khan

Homework No. 01 SOLUTION

Release Date: Jan 31st, 2026 **Due by:** Feb 7th, 2026 11:59 PM

Total points: 100 **Points obtained:**

Student Name:	Student ID:	Section:

Purpose:

The purpose of this assignment is to help you apply the concepts of access control, security attacks, security services, and security mechanisms. In addition, concepts of modular arithmetic and concepts from number theory like extended Euclidean algorithm, linear Diophantine equation, residue matrices will be reinforced.

Instructions:

1. This assignment should be done individually.
2. All questions should be answered in **black ink only**.
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Your assignments will be checked by instructor.
2. You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks.
3. Zero will be given if the assignment is found to be plagiarized.
4. Untidy work will result in a reduction of your points.

Late submission penalty:

- 1-day late submission – 20% deduction of the maximum allowable marks
- 2-days late submission – 40% deduction of the maximum allowable marks
- No submission will be accepted after two days of the original deadline

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	✓
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	

Q1. [20 points] Read the ITU-T Recommendation X.800 (Security Architecture for OSI) document. After reading, write a structured response that explains security attacks, security services, and security mechanisms, and how they relate to each other in X.800.

Your answer must include:

1. Security Attacks
   - Define what X.800 means by *security attacks*.
   - Differentiate between the main categories of attacks (as presented in X.800).
   - Give at least 2 examples of attacks from each category.
2. Security Services vs. Security Mechanisms
   - Define *security services* and *security mechanisms* (in your own words, based on X.800).
   - Write at least 6 main differences between services and mechanisms (e.g., "what" vs "how," abstraction level, design goal vs implementation approach, where they are applied, etc.).
   - Provide at least 3 examples of security services and 3 examples of security mechanisms from X.800.
3. Relationship (Attacks → Services → Mechanisms)
   - Explain (in one paragraph) how security services counter security attacks and how security mechanisms implement/enforce security services.
   - Provide one mapping table (minimum 6 rows) that links:

an attack → the service(s) that help address it → the mechanism(s) that can be used to provide that service.

Answer:
1) Security Attacks (as in X.800)

X.800 frames an attack as the realization of an intentional threat against information/systems—i.e., when a deliberate threat actually happens, it can be considered an "attack."

Main categories of attacks

X.800 distinguishes threats (and therefore attacks) as:

- Passive threats/attacks: unauthorized disclosure without changing system state.
- Active threats/attacks: deliberate unauthorized change to the state of the system. X.800 also lists typical examples of active threats (modification, replay, insertion, masquerade, denial of service).

A) Passive attacks

1. Traffic analysis — inferring information by observing traffic flows (presence/absence, amount, direction, frequency).
2. Message stream observation (eavesdropping/monitoring) — observing data streams to learn information without altering them. X.800 explicitly lists "message stream observation" as something cryptographic protection may address.

B) Active attacks

1. Masquerade — pretending to be a different entity.
2. Replay — reusing previously valid communications (X.800 discusses replay protection/liveness in authentication exchange contexts).
3. Modification of messages / insertion of spurious messages — listed as active threats in X.800.
4. Denial of service — preventing authorized access or delaying time-critical operations.
5. Repudiation — denying having participated in a communication.

2) Security Services vs Security Mechanisms

Security service:
A service provided by an OSI layer that ensures adequate security of systems or data transfers.

Security mechanism:
X.800 describes mechanisms as the means/techniques that implement security services—it explicitly discusses "mechanisms which implement those services," and lists specific mechanisms (encipherment, digital signatures, etc.).

At least 6 main differences

1. "What" vs "How"
   - Service = *what protection is provided* (e.g., confidentiality).
   - Mechanism = *how that protection is achieved* (e.g., encipherment).
2. Abstraction level
   - Services are higher-level capabilities offered to users/layers.
   - Mechanisms are concrete technical methods embedded in layers.
3. Stability vs variability
   - A service goal stays the same (e.g., "data integrity").
   - Multiple mechanisms may be chosen/combined to realize the same service.
4. Relationship to threats
   - Services are defined in terms of countering threats (e.g., integrity counters active threats).
   - Mechanisms are the actual tools used to counter those threats.
5. Where they appear in X.800
   - Services are specified in section 5.2.
   - Specific mechanisms are specified in section 5.3 (and pervasive mechanisms in 5.4).
6. Composability
   - A single mechanism can support multiple services (e.g., encipherment supports confidentiality and may complement other mechanisms).
   - A single service may require multiple mechanisms together (X.800 notes mechanisms can implement combinations of services).

Examples from X.800
Security services (3 examples)

- Peer entity authentication and data origin authentication (under Authentication).
- Access control (protection against unauthorized use of resources).
- Data confidentiality (e.g., connection confidentiality, traffic flow confidentiality).
  (Other valid ones in X.800: data integrity, non-repudiation.)

Security mechanisms (3 examples)

- Encipherment (encryption).
- Digital signature mechanism.
- Traffic padding mechanism (spurious traffic to resist traffic analysis).
  (Other valid ones in X.800: authentication exchange, routing control, notarization, security audit trail, etc.)

3) Relationship (Attacks → Services → Mechanisms)

In X.800, attacks (realized threats) motivate security services (the protections you want), and those services are delivered using security mechanisms (the techniques that implement the protections). For example, traffic analysis is a passive attack; X.800 defines traffic flow confidentiality as a confidentiality service to protect against traffic analysis, and notes that traffic padding (with confidentiality protection) can be used to provide protection against traffic analysis.

Attack (example)	Service(s) that address it	Mechanism(s) that provide the service
Traffic analysis	Traffic flow confidentiality	Traffic padding + confidentiality (encipherment)
Message stream observation/eavesdropping	Data confidentiality (e.g., connection confidentiality)	Encipherment
Masquerade	Peer entity authentication / data origin authentication	Authentication exchange (with replay/liveness protection)
Replay (replay of messages/connection)	Peer entity authentication (with liveness) + data integrity (stream)	Authentication exchange with handshakes/time-stamping; integrity mechanisms (e.g., sequence-related integrity)
Modification of messages / insertion of spurious messages	Data integrity (connection integrity)	Data integrity mechanisms (cryptographic check values / manipulation detection)
Repudiation	Non-repudiation (proof of origin / delivery)	Digital signature and/or notarization mechanisms
Denial of service	Availability (as a security property/service goal)	Detection/logging + recovery actions (audit trail, event detection, security recovery)

Q2. [10 points] Differentiate between Mandatory Access Control (MAC), Discretionary Access Control (DAC), Role-Based Access Control (RBAC), and Attribute-Based Access Control (ABAC).

Your answer must include:

1. Define MAC, DAC, RBAC, and ABAC (2–3 lines each).
2. Write at least 6 key differences among them, focusing on points like:
   - *Who makes the access decision?*

- *How permissions are assigned and managed*
- *Flexibility vs strictness*
- *Scalability in large organizations*
- *Typical use cases*

3. Give one real-world example for each model (e.g., operating systems, organizations, or scenarios).
4. Scenario: A university system includes student records, faculty grading, and finance/payroll data.
   - Choose the most suitable model for each of the three areas (records, grading, payroll) and justify each choice in 2–3 lines.

Answer:

1)

DAC (Discretionary Access Control)
Access is controlled by the owner of an object/resource (e.g., a file owner). The owner can grant or revoke permissions to other users. It is flexible but can be less secure if users share permissions carelessly.

MAC (Mandatory Access Control)
Access is controlled by a central authority/system policy, not by the resource owner. Objects and users have security labels/clearances (e.g., Confidential, Secret), and rules strictly decide access. It is very strong but less flexible.

RBAC (Role-Based Access Control)
Permissions are assigned to roles (e.g., "Instructor", "Student", "Registrar"), and users gain permissions by being assigned roles. It simplifies management in organizations and supports least privilege through well-designed roles.

ABAC (Attribute-Based Access Control)
Access decisions are based on attributes of the user, resource, action, and environment (e.g., user department, resource type, time of day, location, device). Policies are often written as logical rules, making it highly flexible and context-aware.

2) Key differences

1. Who controls permissions
   - DAC: Resource owner decides.
   - MAC: System/central policy decides.
   - RBAC: Admin defines roles; roles define permissions.
   - ABAC: Policy engine evaluates attributes and rules.

2. How access is granted
   - DAC: Direct user-to-resource permissions (ACLs like read/write).
   - MAC: Labels/clearance + enforced rules (e.g., "no read up").
   - RBAC: User → Role → Permissions.
   - ABAC: User/Resource/Context attributes → Policy decision.

3. Flexibility
   - DAC: High flexibility (easy to share).

    - MAC: Low flexibility (strict policy).
    - RBAC: Medium (depends on role design).
    - ABAC: Very high (fine-grained, condition-based).

4. Security strength (typical)
    - DAC: Weaker against insider misuse (owners can over-share).
    - MAC: Strongest for confidentiality in high-security settings.
    - RBAC: Strong if roles are well-designed and reviewed.
    - ABAC: Strong and precise, but depends on correct policy/attributes.

5. Scalability/administration
    - DAC: Gets messy at scale (many ACLs).
    - MAC: Scales for strict environments but is rigid to change.
    - RBAC: Scales well (manage roles instead of individual users).
    - ABAC: Scales well for complex orgs but policy design is harder.

6. Granularity
    - DAC: Usually coarse to medium (file permissions, ACL entries).
    - MAC: Medium (label-based; can be strict but not always detailed).
    - RBAC: Medium (role-level granularity).
    - ABAC: Fine-grained (supports many conditions like time/location).

7. Typical environments
    - DAC: General-purpose OS/file sharing (personal and office use).
    - MAC: Military/government/classified systems, high assurance.
    - RBAC: Enterprises, universities, hospitals, ERP systems.
    - ABAC: Cloud/IAM, zero trust environments, large distributed orgs.

3) One real-world example for each

- DAC example: In Linux/Windows file permissions, the file owner can share a file and set read/write permissions for others.
- MAC example: SELinux (Linux security module) enforces centrally defined policies; users cannot bypass policy just because they "own" a file.
- RBAC example: In a university LMS, "Student" can view materials, "Instructor" can upload and grade, "TA" can grade but not change course settings.
- ABAC example: In cloud IAM (e.g., access to a database allowed only if department=Finance AND device is compliant AND request is during office hours).

4) University scenario: best model for each area

A) Student Records (transcripts, grades history, personal info)
Best choice: RBAC (often with ABAC on top).
Reason: Many users fit stable job functions (Student, Advisor, Registrar). RBAC makes it easy to ensure only authorized roles can access records. ABAC can add rules like "advisor can view records only of assigned advisees."

B) Faculty Grading (enter grades, modify marks, view submissions)

Best choice: RBAC.
Reason: Roles map cleanly: Instructor/TA can grade; Student can submit and view own grades; Department admin can audit. RBAC prevents random sharing that DAC might allow.

C) Finance/Payroll (salaries, bank details, tax records)
Best choice: MAC or ABAC (depending on organization maturity).
- MAC is ideal if strict confidentiality is required with centralized enforcement (cannot be overridden by owners).
- ABAC is ideal if you need fine-grained rules like "Payroll staff can access only payroll data for their campus/unit, only from secure network, only during work hours."

Final pick:
- Student records → RBAC
- Grading → RBAC
- Payroll → ABAC (or MAC for maximum strictness)

Q3. [10 points] Using the Euclidean algorithm, find the greatest common divisor of the following pair of integers.
- 300 and 61

Based on your result, also comment whether the pair is relatively-prime or not relatively-prime.

Answer:
Find gcd(300, 61):

q	r_1	r_2	r
4	300	61	56
1	61	56	5
11	56	5	1
5	5	1	0
—	**1**	**0**	—

Table 1: Euclidean Algorithm for gcd(300, 61)

- Result: gcd(300, 61) = 1.
- Conclusion: Since the gcd = 1, the pair is Relatively-Prime.

Q4. [10 points] Using the extended Euclidean algorithm, find the greatest common divisor of the following pair and the value of s and t.
- 400 and 40

Answer:
Find gcd(400, 40) and values of s and t:

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
10	400	40	0	1	0	1	0	1	-10
—	**40**	**0**	—	**0**	**1**	—	**1**	**-10**	—

Table 2: Extended Euclidean Algorithm for gcd(400, 40)

- Result: The gcd(400, 40) is 40.
- Coefficients: s = 0 and t = 1.
    - *Check:* 400(0) + 40(1) = 40.

Q5. [10 points] Find the results of the following operations:

- 23 mod 7
- 100 mod 10
- -87 mod 13
- 15 mod 15
- 244 mod 26

Answer:

- 23 mod 7 = 2 (23 = 3 x 7 + 2)
- 100 mod 10 = 0
- -87 mod 13 ⟹ -87 = (-7 x 13) + 4 = 4
- 15 mod 15 = 0
- 244 mod 26 = 10 (244 = 9 x 26 + 10)

Q6. [5 points] Perform the following operations using reduction first.

- (50223 + 10372) mod 10
- (244 × 332) mod 10

Answer:

- (50223 + 10372) mod 10 = (3 + 2) mod 10 = 5
- (244 x 332) mod 10 = (4 x 2) mod 10 = 8

Q7. [5 points] Find the multiplicative inverse of the following integer in Z_{26} using the extended Euclidean algorithm.

- 11

Answer:

We keep:

$$r = r_1 - qr_2, \quad t = t_1 - qt_2$$

Initialize:

- $r_1 = 26,\ r_2 = 11$
- $t_1 = 0,\ t_2 = 1$

Extended Euclidean Table (without s-columns)

q	r_1	r_2	r	t_1	t_2	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
—	1	0	—	-7	26	—

Inverse from the row where $r_1 = 1$

At the row $r_1 = 1$, we have $t_1 = -7$.
That means:

$$11 \cdot (-7) \equiv 1 \pmod{26}$$

So:

$$11^{-1} \equiv -7 \equiv 19 \pmod{26}$$

✅ **Multiplicative inverse of 11 in $\mathbb{Z}_{26}$ is $\boxed{19}$.**

Quick check:

$$11 \times 19 = 209 = 26 \cdot 8 + 1 \Rightarrow 209 \equiv 1 \pmod{26}.$$

Q8: [5 points] A post office sells only 39-cent and 15-cent stamps. Find the number of stamps a customer needs to buy to put \$2.70 postage on a package. Find a few solutions.

Answer:

Let

- x = number of **39-cent** stamps
- y = number of **15-cent** stamps

Total postage is 270 cents:

$$39x + 15y = 270$$

Step 1: Compute $d = \gcd(39, 15)$ and reduce

$$\gcd(39, 15) = 3$$

Since $3 \mid 270$, solutions exist.

Divide by 3:

$$13x + 5y = 90$$

So $a_1 = 13,\ b_1 = 5,\ c_1 = 90$.

Step 2: Find s, t such that $13s + 5t = 1$ using a table with s_1, s_2, s and t_1, t_2, t

We use:

$$q = \left\lfloor \frac{r_1}{r_2} \right\rfloor, \quad r = r_1 - qr_2, \quad s = s_1 - qs_2, \quad t = t_1 - qt_2$$

Initialize:

- $r_1 = 13,\ r_2 = 5$
- $s_1 = 1,\ s_2 = 0$
- $t_1 = 0,\ t_2 = 1$

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
2	13	5	3	1	0	1	0	1	-2
1	5	3	2	0	1	-1	1	-2	3
1	3	2	1	1	-1	2	-2	3	-5
2	2	1	0	-1	2	-5	3	-5	13
—	1	0	—	2	-5	—	-5	13	—

From the last row where $r_1 = 1$, we read:

$$s = 2, \quad t = -5$$

So:

$$13(2) + 5(-5) = 26 - 25 = 1$$

Step 3: Particular solution (using the formula)

From the method:

$$x_0 = \left(\frac{c}{d}\right) s, \qquad y_0 = \left(\frac{c}{d}\right) t$$

Here $\frac{c}{d} = \frac{270}{3} = 90$, so:

$$x_0 = 90 \cdot 2 = 180, \qquad y_0 = 90 \cdot (-5) = -450$$

A particular integer solution is:

$$(x_0, y_0) = (180, -450)$$

Step 4: General solutions

Using:

$$x = x_0 + k\left(\frac{b}{d}\right), \qquad y = y_0 - k\left(\frac{a}{d}\right)$$

$$\frac{b}{d} = \frac{15}{3} = 5, \qquad \frac{a}{d} = \frac{39}{3} = 13$$

So:

$$x = 180 + 5k$$

$$y = -450 - 13k$$

where $k \in \mathbb{Z}$.

Step 5: Find valid stamp solutions ($x \geq 0,\ y \geq 0$)

From $x \geq 0$

$$180 + 5k \geq 0 \Rightarrow k \geq -36$$

From $y \geq 0$

$$-450 - 13k \geq 0 \Rightarrow -13k \geq 450 \Rightarrow k \leq -35$$

So:

$$-36 \leq k \leq -35$$

Hence $k = -36$ or $k = -35$.

If $k = -36$

$$x = 180 + 5(-36) = 0, \quad y = -450 - 13(-36) = 18$$

$$(x, y) = (0, 18)$$

If $k = -35$

$$x = 180 + 5(-35) = 5, \quad y = -450 - 13(-35) = 5$$

$$(x, y) = (5, 5)$$

Final answers (a few solutions)

Valid nonnegative solutions are:

1. **0** stamps of 39¢ and **18** stamps of 15¢ $\Rightarrow$ total stamps $= 18$
2. **5** stamps of 39¢ and **5** stamps of 15¢ $\Rightarrow$ total stamps $= 10$

These are the **only** nonnegative integer solutions.

Q9: [5 points] Find all solutions to the following linear equation:

- 3x + 4 ≡ 6 (mod 13)

Answer:

Step 1: Convert to the form $ax \equiv b \pmod{n}$

Subtract 4 from both sides (i.e., add the additive inverse of 4 mod 13, which is -4):

$$3x \equiv 6 - 4 \pmod{13}$$

$$3x \equiv 2 \pmod{13}$$

Now it is in the form $ax \equiv b \pmod{n}$ with $a = 3,\ b = 2,\ n = 13$.

Step 2: Check $\gcd(a, n)$

$$d = \gcd(3, 13) = 1$$

Since $d = 1$, a unique solution mod 13 exists.

Step 3: Multiply by the multiplicative inverse of a (mod n)

Find $3^{-1} \pmod{13}$.
We need:

$$3u \equiv 1 \pmod{13}$$

Try $u = 9$:

$$3 \cdot 9 = 27 \equiv 1 \pmod{13} \quad (\text{since } 27 = 13 \cdot 2 + 1)$$

So:

$$3^{-1} \equiv 9 \pmod{13}$$

Multiply both sides of $3x \equiv 2 \pmod{13}$ by 9:

$$x \equiv 9 \cdot 2 \pmod{13}$$

$$x \equiv 18 \pmod{13}$$

$$x \equiv 5 \pmod{13}$$

So the particular solution is $x_0 = 5$.

Step 4: Write all solutions

Because $d = 1$, there is only **one** solution modulo 13. Therefore:

$$\boxed{x \equiv 5 \pmod{13}}$$

Verification

$$3(5) + 4 = 15 + 4 = 19 \equiv 6 \pmod{13} \quad (\text{since } 19 = 13 + 6)$$

Q10: [10 points] Find the determinant and the multiplicative inverse of each residue matrix over Z_{26}.

a. $\begin{bmatrix} 1 & 5 \\ 3 & 4 \end{bmatrix}$

b. $\begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$

Answer:

We work in $\mathbb{Z}_{26}$. A matrix is invertible mod 26 **iff** $\gcd(\det(A), 26) = 1$.

(a) $A = \begin{bmatrix} 1 & 5 \\ 3 & 4 \end{bmatrix}$

Determinant

$$\det(A) = (1)(4) - (5)(3) = 4 - 15 = -11 \equiv 15 \pmod{26}$$

$\gcd(15, 26) = 1$ so A is invertible.

Inverse

For a 2×2 matrix, $A^{-1} \equiv (\det A)^{-1} \operatorname{adj}(A) \pmod{26}$, where

$$\operatorname{adj}(A) = \begin{bmatrix} 4 & -5 \\ -3 & 1 \end{bmatrix} \equiv \begin{bmatrix} 4 & 21 \\ 23 & 1 \end{bmatrix} \pmod{26}$$

Also,

$$15^{-1} \pmod{26} = 7 \quad (\text{since } 15 \cdot 7 = 105 \equiv 1 \pmod{26})$$

So

$$A^{-1} \equiv 7 \begin{bmatrix} 4 & 21 \\ 23 & 1 \end{bmatrix} \equiv \begin{bmatrix} 2 & 17 \\ 5 & 7 \end{bmatrix} \pmod{26}$$

Answer (a):

$$\det(A) \equiv 15 \pmod{26}, \qquad A^{-1} \equiv \begin{bmatrix} 2 & 17 \\ 5 & 7 \end{bmatrix} \pmod{26}$$

(b) $B = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$

Determinant (value mod 26)

$$\det(B) = -939 \equiv 23 \pmod{26}$$

$\gcd(23, 26) = 1$ so B is invertible.

Inverse

$$B^{-1} \equiv (\det B)^{-1}\,\mathrm{adj}(B) \pmod{26}$$

First,

$$23^{-1} \pmod{26} = 17 \quad (\text{since } 23 \cdot 17 = 391 \equiv 1 \pmod{26})$$

Also (cofactor/adjugate computed and reduced mod 26):

$$\mathrm{adj}(B) \equiv \begin{bmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{bmatrix} \pmod{26}$$

Therefore,

$$B^{-1} \equiv 17 \begin{bmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{bmatrix} \equiv \begin{bmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{bmatrix} \pmod{26}$$

Answer (b):

$$\det(B) \equiv 23 \pmod{26}, \qquad B^{-1} \equiv \begin{bmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{bmatrix} \pmod{26}$$

Q11: [10 points] Find all solutions to the following sets of linear equations:

a. $x + 2y + 3z \equiv 4 \pmod{16}$
   $y + 4z \equiv 9 \pmod{16}$
   $5x + 6y + 14z \equiv 3 \pmod{16}$

b. $x + 2y + 3z \equiv 2 \pmod{16}$
   $y + 4z \equiv 1 \pmod{16}$
   $5x + 6y \equiv 13 \pmod{16}$

Answer:

a.

Matrix form $AX \equiv B \pmod{16}$:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 14 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 9 \\ 3 \end{bmatrix}$$

Step 1: Determinant $\det(A)$

Expand along row 1:

$$\det(A) = 1\begin{vmatrix}1 & 4\\ 6 & 14\end{vmatrix} - 2\begin{vmatrix}0 & 4\\ 5 & 14\end{vmatrix} + 3\begin{vmatrix}0 & 1\\ 5 & 6\end{vmatrix}$$

Compute minors:

- $\begin{vmatrix}1 & 4\\ 6 & 14\end{vmatrix} = 1\cdot 14 - 4\cdot 6 = 14 - 24 = -10$
- $\begin{vmatrix}0 & 4\\ 5 & 14\end{vmatrix} = 0\cdot 14 - 4\cdot 5 = -20$
- $\begin{vmatrix}0 & 1\\ 5 & 6\end{vmatrix} = 0\cdot 6 - 1\cdot 5 = -5$

So:

$$\det(A) = 1(-10) - 2(-20) + 3(-5) = -10 + 40 - 15 = 15$$

$$\boxed{\det(A) \equiv 15 \pmod{16}}$$

Since $\gcd(15, 16) = 1$, inverse exists ⇒ **unique solution**.

Step 2: Cofactor matrix C

Cofactor $C_{ij} = (-1)^{i+j} M_{ij}$.

Row 1

- $C_{11} = +\begin{vmatrix}1 & 4\\ 6 & 14\end{vmatrix} = -10$
- $C_{12} = -\begin{vmatrix}0 & 4\\ 5 & 14\end{vmatrix} = -(-20) = 20$
- $C_{13} = +\begin{vmatrix}0 & 1\\ 5 & 6\end{vmatrix} = -5$

Row 2

- $C_{21} = -\begin{vmatrix}2 & 3\\ 6 & 14\end{vmatrix} = -(2\cdot 14 - 3\cdot 6) = -(28 - 18) = -10$
- $C_{22} = +\begin{vmatrix}1 & 3\\ 5 & 14\end{vmatrix} = 14 - 15 = -1$
- $C_{23} = -\begin{vmatrix}1 & 2\\ 5 & 6\end{vmatrix} = -(6 - 10) = 4$

Row 3

- $C_{31} = +\begin{vmatrix}2 & 3\\ 1 & 4\end{vmatrix} = 8 - 3 = 5$
- $C_{32} = -\begin{vmatrix}1 & 3\\ 0 & 4\end{vmatrix} = -(4 - 0) = -4$
- $C_{33} = +\begin{vmatrix}1 & 2\\ 0 & 1\end{vmatrix} = 1$

$$C = \begin{bmatrix}-10 & 20 & -5\\ -10 & -1 & 4\\ 5 & -4 & 1\end{bmatrix}$$

Step 3: Adjoint $\mathrm{adj}(A) = C^T$

$$\mathrm{adj}(A) = \begin{bmatrix} -10 & -10 & 5 \\ 20 & -1 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Reduce mod 16:

- $-10 \equiv 6$, $20 \equiv 4$, $-1 \equiv 15$, $-4 \equiv 12$, $-5 \equiv 11$

$$\mathrm{adj}(A) \equiv \begin{bmatrix} 6 & 6 & 5 \\ 4 & 15 & 12 \\ 11 & 4 & 1 \end{bmatrix} \pmod{16}$$

Step 4: Compute $(\det A)^{-1} \pmod{16}$

$$15^{-1} \equiv 15 \pmod{16} \quad (\text{since } 15 \cdot 15 = 225 \equiv 1)$$

So:

$$A^{-1} \equiv 15 \cdot \mathrm{adj}(A) \pmod{16}$$

Multiplying by 15 is the same as negating mod 16 (because $15 \equiv -1$):

$$A^{-1} \equiv -\,\mathrm{adj}(A) \equiv \begin{bmatrix} 10 & 10 & 11 \\ 12 & 1 & 4 \\ 5 & 12 & 15 \end{bmatrix} \pmod{16}$$

Step 5: Solve $X \equiv A^{-1}B \pmod{16}$

$$X \equiv \begin{bmatrix} 10 & 10 & 11 \\ 12 & 1 & 4 \\ 5 & 12 & 15 \end{bmatrix} \begin{bmatrix} 4 \\ 9 \\ 3 \end{bmatrix} \pmod{16}$$

x

$$x \equiv 10 \cdot 4 + 10 \cdot 9 + 11 \cdot 3 = 40 + 90 + 33 = 163 \equiv 3$$

y

$$y \equiv 12 \cdot 4 + 1 \cdot 9 + 4 \cdot 3 = 48 + 9 + 12 = 69 \equiv 5$$

z

$$z \equiv 5 \cdot 4 + 12 \cdot 9 + 15 \cdot 3 = 20 + 108 + 45 = 173 \equiv 13$$

✅ Final answer (unique mod 16)

$$\boxed{x \equiv 3, \quad y \equiv 5, \quad z \equiv 13 \pmod{16}}$$

Quick verification

1. $x + 2y + 3z = 3 + 10 + 39 = 52 \equiv 4$ ✅
2. $y + 4z = 5 + 52 = 57 \equiv 9$ ✅
3. $5x + 6y + 14z = 15 + 30 + 182 = 227 \equiv 3$ ✅

b.

Matrix form $AX \equiv B \pmod{16}$:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 1 \\ 13 \end{bmatrix}$$

Step 1: Compute $\det(A)$

Expand along the first row:

$$\det(A) = 1\begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} - 2\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} + 3\begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix}$$

Compute each 2×2:

- $\begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} = 1 \cdot 0 - 4 \cdot 6 = -24$
- $\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} = 0 \cdot 0 - 4 \cdot 5 = -20$
- $\begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = 0 \cdot 6 - 1 \cdot 5 = -5$

So:

$$\det(A) = 1(-24) - 2(-20) + 3(-5) = -24 + 40 - 15 = 1$$

$$\boxed{\det(A) = 1 \equiv 1 \pmod{16}}$$

Since $\gcd(1, 16) = 1$, the inverse exists and the solution is **unique mod 16**.

Step 2: Find the cofactor matrix C

Cofactor $C_{ij} = (-1)^{i+j} M_{ij}$.

Row 1

- $C_{11} = +\begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} = -24$
- $C_{12} = -\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} = -(-20) = 20$
- $C_{13} = +\begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = -5$

Row 2

- $C_{21} = -\begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} = -(2 \cdot 0 - 3 \cdot 6) = -(-18) = 18$
- $C_{22} = +\begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} = 1 \cdot 0 - 3 \cdot 5 = -15$
- $C_{23} = -\begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} = -(1 \cdot 6 - 2 \cdot 5) = -(6 - 10) = 4$

Row 3

- $C_{31} = +\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 2 \cdot 4 - 3 \cdot 1 = 8 - 3 = 5$
- $C_{32} = -\begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} = -(1 \cdot 4 - 3 \cdot 0) = -4$
- $C_{33} = +\begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 - 2 \cdot 0 = 1$

So:

$$C = \begin{bmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{bmatrix}$$

Step 3: Compute $\mathrm{adj}(A) = C^T$

$$\mathrm{adj}(A) = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Reduce entries mod 16:

- $-24 \equiv 8$
- $18 \equiv 2$
- $20 \equiv 4$
- $-15 \equiv 1$
- $-4 \equiv 12$
- $-5 \equiv 11$

So:

$$\mathrm{adj}(A) \equiv \begin{bmatrix} 8 & 2 & 5 \\ 4 & 1 & 12 \\ 11 & 4 & 1 \end{bmatrix} \pmod{16}$$

Step 4: Compute $A^{-1} \pmod{16}$

$$A^{-1} \equiv (\det A)^{-1} \operatorname{adj}(A) \pmod{16}$$

But $\det A \equiv 1$, so $(\det A)^{-1} \equiv 1$. Hence:

$$\boxed{A^{-1} \equiv \operatorname{adj}(A) \equiv \begin{bmatrix} 8 & 2 & 5 \\ 4 & 1 & 12 \\ 11 & 4 & 1 \end{bmatrix} \pmod{16}}$$

Step 5: Solve $X \equiv A^{-1}B \pmod{16}$

$$X \equiv \begin{bmatrix} 8 & 2 & 5 \\ 4 & 1 & 12 \\ 11 & 4 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 13 \end{bmatrix} \pmod{16}$$

Compute each component:

x

$$x \equiv 8(2) + 2(1) + 5(13) = 16 + 2 + 65 = 83 \equiv 3 \pmod{16}$$

y

$$y \equiv 4(2) + 1(1) + 12(13) = 8 + 1 + 156 = 165 \equiv 5 \pmod{16}$$

z

$$z \equiv 11(2) + 4(1) + 1(13) = 22 + 4 + 13 = 39 \equiv 7 \pmod{16}$$

✅ Final Answer (unique mod 16)

$$\boxed{x \equiv 3, \quad y \equiv 5, \quad z \equiv 7 \pmod{16}}$$

Quick verification (substitution mod 16)

1. $x + 2y + 3z = 3 + 10 + 21 = 34 \equiv 2$ ✅
2. $y + 4z = 5 + 28 = 33 \equiv 1$ ✅
3. $5x + 6y = 15 + 30 = 45 \equiv 13$ ✅